

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 17-May-2025

Time: Three Hours

Max. Marks: 75

Medicinal Chemistry - I

Q.P. CODE: 5014

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory

LONG ESSAYS

2 x 10 = 20 Marks

1. Define and classify sympathomimetic agents. Give the structure of one drug from each class. Write the synthesis and uses of Phenylephrine and Propranolol

OR

What are anticonvulsants? Classify them with suitable examples. Write the mechanism of action of barbiturates. Outline the synthesis of Phenytoin and Carbamazepine

2. Define NSAIDs. Classify them with examples. Write the structure of one drug from each class. Explain the synthesis of Mefenamic acid

SHORT ESSAYS

7 x 5 = 35 Marks

3. Discuss the SAR of Phenothiazines

OR

Explain the role of protein binding and hydrogen bonding properties on drug action

4. Add a note on Phase II drug biotransformation giving suitable reaction examples

OR

Discuss the SAR of Morphine analogues

5. What are cholinergic blocking agents? Outline the synthesis of Dicyclomine hydrochloride

6. Write the biosynthesis of catecholamines

7. Define and classify general anaesthetic agents with examples. Outline the synthesis of Halothane

8. Discuss the SAR of Benzodiazepines

9. Explain the mechanism of action of NSAIDs

SHORT ANSWERS

10 x 2 = 20 Marks

10. Write the biosynthesis of Acetylcholine
11. Give two examples of oxidative drug metabolism
12. Write a note on adrenergic receptors
13. Write the chemical structure and uses of Phenobarbitone and Paraldehyde
14. Draw the chemical structure of Malathion and Parathion
15. Define Narcotic antagonists, and give examples
16. Outline the synthesis of Ketamine Hydrochloride
17. Write the chemical structure and uses of Tolazoline
18. Classify parasympathomimetic agents with examples
19. Write the structure and uses of any two inhalation anaesthetics
